LDDN: A Lightweight Dual-Domain Directional Network for Remote Sensing Change Detection

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***Abstract*—Remote sensing image change detection holds significant importance in fields such as urban dynamic monitoring and ecological assessment. However, restricted by parameter capacity, lightweight models struggle to capture subtle changes in complex scenes. Furthermore, factors such as slight registration errors and local textural noise often trigger high-frequency pseudo-change responses, leading to false detections. To this end, this article proposes a lightweight dual-domain directional network (LDDN) for resource-constrained scenarios, achieving efficient and reliable change detection through the synergistic design of frequency-domain noise suppression and spatial-domain directional modeling. Specifically, to suppress unreliable high-frequency responses induced by factors like registration errors, a directional wavelet attention module (DWAM) is designed. It utilizes the directional high-frequency subbands of the discrete wavelet transform (DWT) to perform adaptive screening, thereby suppressing unreliable components and mining structural cues associated with real changes. To enhance the capability of lightweight models in capturing weak change targets, a directional spatial fusion module (DSFM) is proposed, which constructs long-range anisotropic modeling capabilities at a low cost by decoupling horizontal and vertical contextual dependencies. Furthermore, to ensure semantic consistency across multiple scales, a multiscale interaction module (MIM) is introduced to facilitate cross-scale synergy between frequency-domain filtering and spatial-domain directional modeling. The experimental results demonstrate that with only 1.98M parameters and 2.58G FLOPs, LDDN achieves competitive or superior detection performance compared to various state-of-the-art methods across four public benchmark datasets. Particularly on the challenging CLCD dataset, the $F1$-score and intersection over union (IoU) are improved by 0.83% and 1.07%, respectively, compared to the second-best method. The code is available at: https://github.com/Vi-chen/LDDN**

***Index Terms*—Change detection, directional modeling, dual-domain learning, lightweight network, remote sensing.**

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I. Introduction

REMOTE sensing change detection (RSCD) aims to automatically locate and identify surface change regions from remote sensing images acquired at different times in the same area, serving as one of the fundamental problems in intelligent remote sensing interpretation [1], [2]. With the rapid development of Earth observation platforms such as high-resolution satellites and uncrewed aerial vehicles (UAVs), the acquisition of bitemporal images has become more convenient, promoting the widespread application of RSCD in scenarios such as urban expansion monitoring [3], disaster damage assessment [4], and land cover analysis [5]. Compared to single-image understanding, RSCD requires characterizing real changes reliably and locating them precisely under conditions where temporal differences coexist with scene complexity. Therefore, it requires the model not only to possess the ability to discriminate changes but also to effectively model the reliability of change responses under complex interference conditions. Furthermore, image quality and representation factors, such as spectral fidelity, noise suppression, and computational efficiency, are also important for reliable remote sensing interpretation and can further affect change detection performance [6], [7]. Recent material-proportion-guided image enhancement studies have shown that physically meaningful priors can improve spatial detail reconstruction and material representation in remote sensing imagery [8].

Early RSCD approaches mainly relied on algebraic and statistical techniques to measure bitemporal differences [9]. Algebra-based methods perform pixelwise change quantification, such as change vector analysis (CVA) [10] and spectral angle mapper (SAM) [11], while transformation-based methods project images into discriminative feature spaces, including principal component analysis (PCA) [12], independent component analysis (ICA) [13], and linear discriminant analysis (LDA) [14]. However, such methods mainly rely on handcrafted features and statistical modeling assumptions, which limit their ability to characterize cross-temporal differences under complex land-cover conditions, thereby affecting the stability of change detection results.

Benefiting from the powerful feature extraction capabilities of convolutional neural networks (CNNs), deep-learning-based methods have gradually replaced traditional algebraic and statistical methods to become mainstream. Early fully convolutional networks (FCNs) [15] achieved pixel-level prediction in an end-to-end manner, and subsequently, Siamese-based dual-stream networks [16], [17], [18], [19] further enhanced

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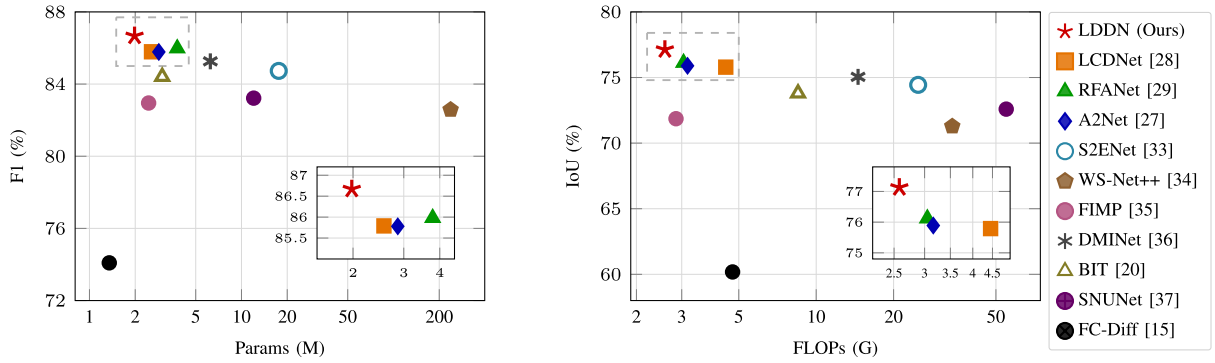


Fig. 1. Performance comparison between the proposed LDDN and other state-of-the-art methods. The metrics ($F1$-score and IoU) represent the average values across four benchmark datasets (SYSU-CD, CLCD, WHU-CD, and LEVIR-CD).

the discriminative ability for bitemporal features. To overcome the limitation of the local receptive field in CNNs, researchers introduced Transformer architectures [20], [21], [22], utilizing self-attention mechanisms to capture global dependencies. However, the quadratic computational complexity of Transformers [23] makes them difficult to apply to high-resolution images. To address efficiency issues, the Mamba architecture based on state space models (SSMs) has recently been introduced into vision tasks [24], [25], [26], successfully achieving long-range modeling with linear complexity. Although the aforementioned methods continue to achieve performance breakthroughs, their high dependence on computing power and storage resources poses significant challenges for achieving stable and reliable change detection in resource-constrained scenarios.

To break the computational bottleneck, recent research has been dedicated to exploring efficient lightweight architectures, attempting to achieve precise inference on resource-constrained devices through streamlined feature aggregation strategies. Representative works such as A2Net [27], LCDNet [28], and RFANet [29], [30] have successfully maintained superior performance with extremely low parameter counts by introducing neighborhood aggregation or temporal interaction mechanisms. Although these lightweight methods have achieved breakthroughs in efficiency, they still face a severe tradeoff between "lightweight" and "robustness" in complex high-resolution scenes, specifically reflected in two essential defects. On the one hand, local inconsistencies across phases often manifest as high-frequency pseudo-changes. Since lightweight networks typically reduce the redundancy of feature filtering to compress parameters [31], they struggle to automatically suppress these high-frequency interferences in deep features like large models, leading to numerous false detections. On the other hand, to pursue extreme computational efficiency, lightweight backbones widely adopt isotropic depthwise separable convolutions. However, remote sensing objects (such as roads and boundaries) often possess significant directionality. These isotropic local operators limit the model's ability to capture anisotropic geometric features [32], resulting in fragmented and discontinuous detection results in low-contrast scenes.

Based on the above analysis, this article proposes a lightweight dual-domain directional network (LDDN) for resource-constrained RSCD, where directional perception is employed as a natural cue to connect frequency-domain reliability modeling with spatial-domain geometric representation. In the frequency domain, wavelet-decomposed directional subbands contain structural boundary cues, but they may also include unreliable high-frequency responses caused by misregistration and local textural noise [38], thus necessitating direction-aware screening to isolate reliable cues. Meanwhile, in the spatial domain, prominent remote sensing targets, such as roads and building contours, often exhibit pronounced anisotropic geometries, making orientation-specific modeling beneficial for preserving structural continuity. Since isolated spatial modeling is vulnerable to spurious high-frequency variations [39], and independent frequency filtering is insufficient for maintaining long-range structural continuity [33], a unified directional approach provides a systematic mechanism to leverage their complementary properties. Although multiscale convolutions are effective in enlarging receptive fields [40], they mainly focus on spatial-scale aggregation and are less explicit in coordinating frequency-domain reliability cues with directional spatial dependencies. To instantiate our unified concept, LDDN deploys the directional wavelet attention module (DWAM) for high-frequency noise suppression and the directional spatial fusion module (DSFM) to enhance weak signal perception through long-range anisotropic spatial modeling. Finally, the multiscale interaction module (MIM) is introduced to coordinate these dual-domain directional representations across multiple scales. This coordinated mechanism facilitates cross-scale synergy among the components, thereby improving detection robustness and efficiency compared with conventional single-domain designs. As shown in Fig. 1, the proposed LDDN achieves a good balance between detection accuracy and model efficiency.

The main contributions of this work are summarized as follows.

1) To address the dilemma between detection accuracy and computational efficiency in resource-constrained scenarios, a LDDN is proposed. By synergizing frequency-domain directional denoising and spatial directional modeling, reliable change detection is achieved using only 1.98M parameters and 2.58G FLOPs.
2) To suppress high-frequency pseudo-changes and capture subtle anisotropic changes, a DWAM and a DSFM are designed. DWAM adaptively filters unreliable high-frequency components and mines structural features of

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real changes in the frequency domain, while DSFM decouples spatial context into orthogonal directions to enhance sensitivity to weak changes, achieving this goal with low computational cost.

3) To bridge the semantic gap between frequency-domain reliability screening and spatial directional modeling at different scales, a MIM is proposed. It coordinates dual-domain features within multiscale receptive fields to enhance semantic consistency and further refine change boundaries.

The rest of this article is structured as follows. A brief review of related work is provided in Section II. The overall architecture and specific details of the proposed LDDN are introduced in Section III. The experimental analysis is then presented in Section IV. Finally, a concise conclusion is provided in Section V.

II. Related Works

A. Deep-Learning-Based Change Detection Methods

Early deep-learning-based RSCD methods primarily relied on FCNs and Siamese architectures. FC-Siam [15] established the basic framework for deep change detection by constructing a weight-sharing encoder to extract bitemporal features. Subsequently, to enhance feature discriminability, multiscale feature aggregation and attention mechanisms were widely introduced. For instance, SNUNet [37] alleviated the loss of deep feature information through dense connections, while AFPF-Net [17] adopted a progressive fusion strategy to capture fine-grained difference features. Although recent Transformer [22] and Mamba [26] architectures have refreshed performance benchmarks with their powerful global modeling capabilities, their quadratic or heavy linear computational complexity limits their application in resource-constrained scenarios. Therefore, exploring efficient CNN architectures remains of great significance.

Beyond fully supervised settings, recent studies have also explored weakly supervised and semi-supervised CNN-based paradigms to alleviate the reliance on dense annotations in RSCD. For example, Wu et al. [2] proposed a unified fully convolutional framework for unsupervised, weakly supervised, and regionally supervised change detection, while WS-Net++ [34] introduced a wavelet Siamese network with semi-supervised domain adaptation to reduce cross-domain appearance discrepancies. At the methodological level, label-efficient learning paradigms such as Confucius tri-learning [41], and its subsequent theoretical justification from mutual guide to Confucius tri-learning [42] further highlight the potential of improving feature learning under limited-label settings. These studies mainly emphasize annotation efficiency and data utilization, whereas our work focuses on lightweight yet robust change representation under resource-constrained scenarios by jointly modeling directional cues in the frequency and spatial domains.

In CNN architectures, the geometric morphology of convolution operators is crucial for change modeling. However, existing methods mostly employ standard square convolution kernels. This isotropic design often faces a dilemma when modeling long-range directional targets common in remote sensing images: small kernels suffer from insufficient receptive fields, while large kernels introduce a significant amount of irrelevant background noise. Although strip convolution and strip pooling [43] have shown effectiveness in capturing anisotropic features in semantic segmentation and object detection, their potential for lightweight RSCD remains insufficiently explored. For example, MSPNet [44] improves the structural integrity of road extraction through multiscale strip pooling, while Strip R-CNN [32] employs large-kernel strip convolutions to enhance the detection of objects with large aspect ratios. However, these methods mainly apply strip operations as generic spatial operators for single-image representation. In contrast, the proposed DSFM adapts directional modeling to the bitemporal feature fusion process in lightweight change detection. It decouples the fused change representation into horizontal and vertical directional dependencies and introduces adaptive directional weighting to balance their contributions. This design enables the model to capture long-range anisotropic structures and enhance weak or elongated change regions with low computational overhead.

B. Frequency-Domain-Based Change Detection Methods

Imaging discrepancies in remote sensing imagery, such as seasonal variations and illumination inconsistency, are typically reflected in low-frequency components as global intensity shifts, whereas the structural information of key objects, including buildings and roads, is mainly concentrated in high-frequency components [33], [35], [45]. Relying solely on spatial-domain feature learning makes it difficult to effectively disentangle these interferences, motivating the introduction of frequency-domain awareness to improve model robustness.

Some studies adopt Fourier transform-based analysis to model bitemporal features in the spectral domain. For example, MFGFNet [45] captures contextual information through global frequency-domain filtering to mitigate appearance discrepancies, while FIMP [35] designs a Fourier feature interaction module to adaptively filter amplitude spectrum differences. However, the Fourier transform exhibits a global nature and lacks time–frequency localization capability. As a result, while suppressing global disturbances, it may also weaken local subtle change responses, adversely affecting precise boundary localization.

To compensate for the spatial localization deficiencies of frequency-domain methods, the discrete wavelet transform (DWT) has been introduced into change detection tasks due to its good time–frequency localization properties [46]. HFA-Net [38] utilizes high-frequency components to enhance building edge representation, WS-Net++ [34] constructs a spatial–frequency interaction mechanism to improve boundary integrity, and S2ENet [33] further analyzes high-frequency subbands in shallow features to assist edge alignment. Different from existing wavelet-based methods that mainly use frequency-domain components for feature enhancement or spatial–frequency fusion [47], [48], DWAM focuses on the reliability of directional high-frequency responses. It performs direction-aware screening on wavelet subbands and further reconstructs reliable structural cues through cross-subband

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interaction, aiming to suppress pseudo-changes while preserving change-related boundaries.

C. Lightweight Backbone-Based Change Detection Methods

To accommodate limited computational and storage resources in scenarios such as disaster emergency response and UAV inspection, lightweight change detection aims to maintain competitive accuracy under strict computational budgets. These methods typically employ parameter-efficient backbones, such as MobileNet [31] or EfficientNet [49], as encoders and focus on designing compact feature aggregation strategies. One representative line of work emphasizes efficient feature interaction to compensate for the limited representational capacity of lightweight backbones. For instance, A2Net [27] proposes neighborhood aggregation and progressive change recognition modules, reusing adjacent-level features to enhance temporal representation. LCDNet [28] introduces temporal interaction and gating mechanisms to dynamically fuse bitemporal features with minimal computational cost. To further strengthen weak semantic features, RFANet [29] designs feature enhancement modules and a semantic segmentation–aggregation strategy to guide the model's focus toward change regions. Another line of work pursues extreme architectural compression, such as TinyCD [30], which adopts a Siamese U-Net structure with hybrid attention mask blocks and achieves competitive performance with extremely low parameter counts.

Despite the progress in computational efficiency, most existing lightweight change detection methods are still dominated by a single spatial-domain modeling paradigm. To go beyond this limitation, this work introduces a dual-domain directional modeling paradigm that couples frequency-domain reliability screening with spatial-domain geometric representation under lightweight constraints. Frequency-domain directional modeling emphasizes the reliability of local high-frequency structural responses, whereas spatial-domain directional modeling enhances long-range anisotropic geometric continuity. Since remote sensing changes involve both fine boundaries and larger structural regions, MIM is designed as a cross-scale coordinator to integrate dual-domain directional features under multiple receptive fields, rather than serving as a generic multiscale aggregation block. This design helps form more consistent and structurally complete change representations.

III. Proposed Method

This section details the proposed LDDN. First, the overall architecture of the network is outlined. Subsequently, the DWAM, the DSFM, and the MIM are described in detail. Finally, the loss function used for model optimization is presented.

A. Overview

As illustrated in Fig. 2, the proposed LDDN adopts a Siamese encoder–decoder architecture to achieve accurate and efficient change detection in complex scenarios. The network consists of four key components: a weight-sharing backbone network, a frequency-domain reliability screening stage based on the DWAM, a spatiotemporal fusion stage based on the DSFM, and a feature refinement stage based on the MIM.

Given a pair of bitemporal high-resolution remote sensing images, denoted as $\{I^1, I^2\} \in \mathbb{R}^{H\times W\times 3}$, where H represents the height of the image, W represents the width of the image, and 3 represents the band of the image, LDDN employs the first five stages of a weight-sharing EfficientNet-B4 [49] as the encoder. The outputs of stages 2–5 are selected as multiscale feature representations. For simplicity, these four stages are reindexed, and the corresponding bitemporal features are denoted as $\{F_i^1, F_i^2\}_{i=1}^4 \in \mathbb{R}^{H/2^i\times W/2^i\times C_i}$, where $H/2^i$, $W/2^i$, and C_i represent the spatial resolution and channel dimension of the ith feature level. In change detection tasks, cross-temporal feature comparison often generates unreliable high-frequency difference responses, thereby interfering with change discrimination. To this end, the extracted bitemporal features are first fed into the DWAM to model the reliability of change responses in the frequency domain. DWAM utilizes the DWT to decompose features into high-frequency subbands with directional attributes and suppresses unreliable high-frequency difference responses through soft thresholding and cross-subband interaction, thereby obtaining more stable feature representations. After completing the reliability screening at the frequency domain level, the DSFM is introduced to enhance the spatial discriminative ability between bitemporal features. Different from direct feature concatenation or differencing operations, DSFM decouples the 2-D spatial context into orthogonal dependencies in horizontal and vertical directions via a built-in directional attention (DA) mechanism and models them, respectively. This design effectively models long-range anisotropic context while maintaining low computational overhead, thereby enhancing the model's discriminative ability for weak change regions and generating fused change feature representations. Subsequently, to bridge the semantic gap between the frequency and spatial domains across different scales, the features are input into the MIM. MIM adopts a multikernel parallel structure to coordinate frequency-domain denoising and spatial-domain directional modeling at different spatial scales, further refining the change feature representation. Finally, the refined multiscale features are fused through a progressively upsampling decoder, and the final binary change map is generated by a classifier.

B. Directional Wavelet Attention Module

In the frequency domain analysis of remote sensing images, high-frequency components mainly reflect local spatial discontinuities and fine-grained structural changes. These components contain both structural boundary cues related to real changes and unreliable high-frequency responses caused by registration errors or local texture noise. Based on this analysis, we propose the DWAM, as shown in Fig. 2. It aims to adaptively screen change cues within the DWT domain according to the directional characteristics of subbands, recover structural details through cross-subband interaction, and finally generate a gating mask via hybrid channel attention (HCA) to modulate bitemporal features. For simplicity, we omit the subscript i when the context is clear.

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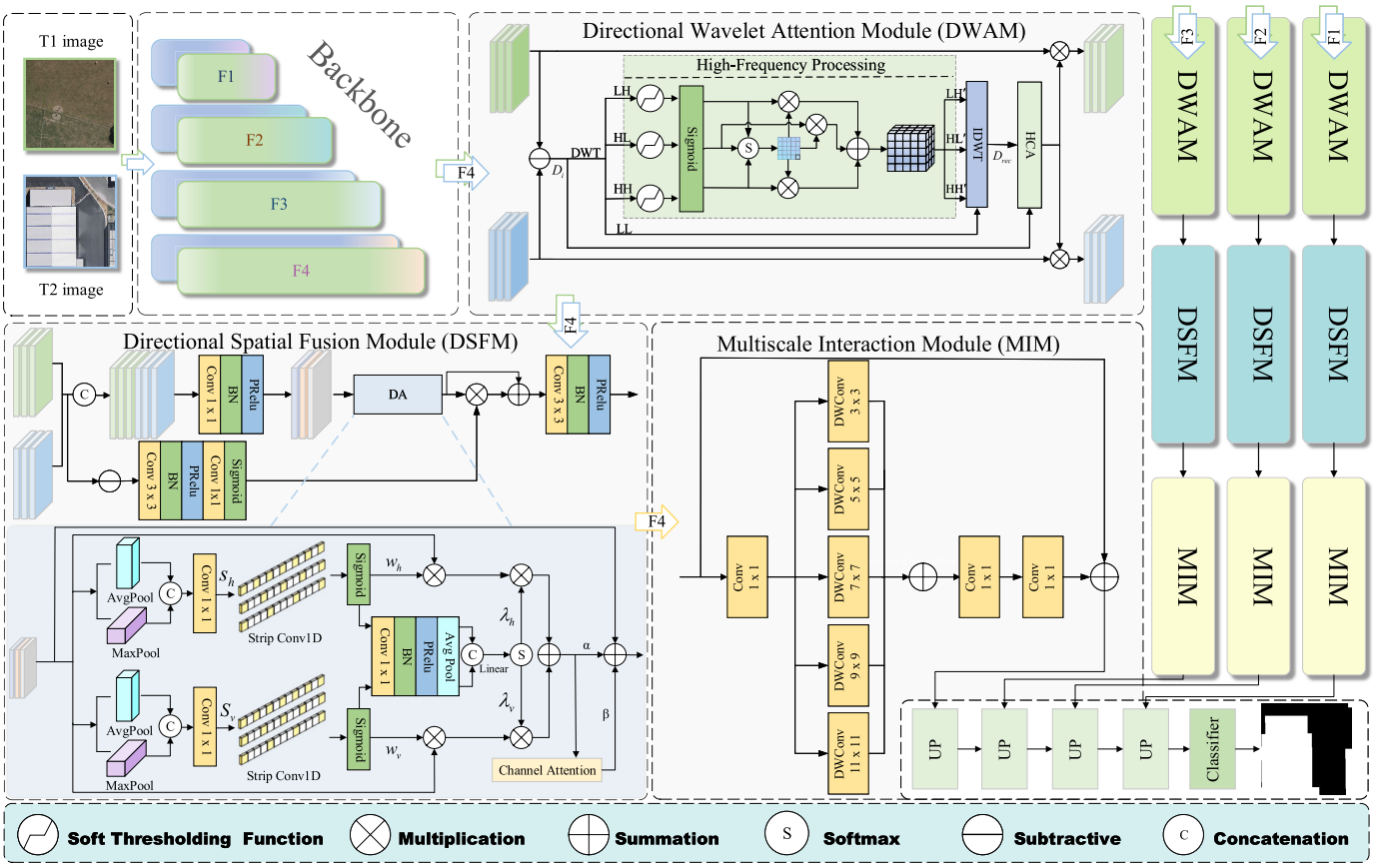


Fig. 2. Framework of LDDN mainly consists of five components: a weight-sharing backbone, a DWAM, a DSFM, a MIM, and a decoder. In LDDN, the first five stages of a pretrained EfficientNet-B4 are utilized as the backbone to extract multiscale features (selecting outputs from stages 2–5). Unreliable high-frequency responses are adaptively screened and suppressed in the frequency domain via DWAM to purify features. Spatial context is decoupled via DSFM to capture weak change features, and semantic consistency across scales is coordinated using MIM. Finally, these features are fed into a progressive upsampling decoder to produce the binary change detection map.

Let the bitemporal features output by the backbone network be denoted as $\{F^1, F^2\}$. First, we compute the difference feature $D_i = F^1 - F^2$ as the internal input of the module. To separate structural information, DWT is utilized to decompose D_i into a low-frequency component LL and three high-frequency directional subbands $\mathcal{W} = \{LH, HL, HH\}$, where LH, HL, and HH represent the vertical, horizontal, and diagonal high-frequency subbands, respectively. Considering that high-frequency subbands contain both structural information and unreliable responses, we design a direction-aware soft thresholding function to obtain reliability-enhanced high-frequency representations $\hat{D}_k$

$$\hat{D}_k = \sigma(G_k) \cdot S_{\text{soft}}(k; \theta_k, \mu_k), \quad k \in \mathcal{W} \tag{1}$$

where $S_{\text{soft}}(\cdot)$ is an adaptive soft-thresholding operation adjusted by parameter $\theta_k \in \mathbb{R}^C$ based on the channel-level mean μ_k, utilized to filter out background noise; $\sigma(\cdot)$ is the Sigmoid activation function based on parameter $G_k \in \mathbb{R}^C$, used to further suppress spurious responses in specific directions.

Since the edges of real ground objects often span across multiple frequency bands, processing each subband independently may destroy structural integrity. To this end, we introduce a cross-subband interaction mechanism. First, a shared high-frequency representation H_{mix} is obtained based on subband energy aggregation, and then it is mapped back to each directional subband through a learnable redistribution matrix $W_{\text{redis}} \in \mathbb{R}^{C\times 3}$

$$D_{\text{rec}} = \text{IDWT}\,(LL, H_{\text{mix}} \cdot W_{\text{redis}}) \tag{2}$$

where IDWT$(\cdot)$ denotes the inverse DWT, and D_{rec} represents the reconstructed difference feature after directional frequency redistribution. This process allows the model to dynamically adjust the energy distribution of different directional components in the frequency domain.

To alleviate the potential semantic loss caused by frequency-domain processing, we design an HCA to refine the difference features, as shown in Fig. 3. We argue that the reconstructed structural feature D_{rec} should serve as a guide to calibrate the original difference feature from two dimensions: the "relative importance" of Softmax and the "absolute activation strength" of the SE module [50]. This process can be expressed as

$$\tilde{D}_i = D_i \cdot \delta(L(\text{GAP}(D_{\text{rec}}))) + \gamma \cdot (D_{\text{rec}} \cdot \text{SE}(D_{\text{rec}})) \tag{3}$$

where $\delta(\cdot)$ denotes the Softmax function, L is a Linear Layer, GAP$(\cdot)$ represents the global average pooling, SE$(\cdot)$ denotes the channelwise attention operation, which models the absolute activation strength of each channel, and γ is a learnable balance coefficient initialized to 0.5, used to

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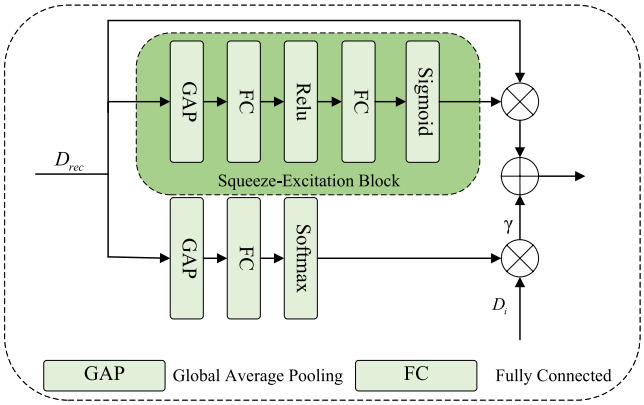


Fig. 3. Illustration of the HCA.

balance the contribution of original difference information and frequency-domain screened information, obtaining the final refined difference feature $\tilde{D}_i$.

Finally, a convolutional layer is used to generate a channel gating map M from $\tilde{D}_i$ to modulate the original bitemporal backbone features, outputting the refined features $\{\tilde{F}^1, \tilde{F}^2\}$

$$\tilde{F}^1 = F^1 \cdot M, \quad \tilde{F}^2 = F^2 \cdot M. \tag{4}$$

Through this process, DWAM effectively reduces the interference of unreliable high-frequency difference responses on change discrimination while maintaining the semantic consistency of bitemporal features.

C. Directional Spatial Fusion Module

To enhance the spatial discriminative ability of bitemporal features while maintaining lightweight computational overhead, we design the DSFM, as shown in Fig. 2. Different from approaches introducing 2-D global attention, DSFM models spatial context via a direction decoupling strategy. It employs strip convolutions to capture long-range spatial dependencies with anisotropic characteristics, thereby improving the lightweight model's discriminative ability for weak change regions.

For the input bitemporal features $\{\tilde{F}^1, \tilde{F}^2\}$, we first generate initial fused features through channel concatenation and convolutional projection, which can be expressed as

$$F_{\text{in}} = \phi\left(\text{Conv}_{1\times1}\left(\text{Cat}\left(\tilde{F}^1, \tilde{F}^2\right)\right)\right) \tag{5}$$

where $\phi(\cdot)$ denotes the parametric rectified linear unit (PReLU) followed by batch normalization, and Cat$(\cdot)$ represents the channelwise concatenation.

Subsequently, DSFM feeds the fused feature into the DA unit. Taking the vertical branch as an example, average pooling and max pooling are first performed on F_{in} along the horizontal dimension. The pooled features are concatenated and compressed via a 1×1 convolution to aggregate 2-D features into a 1-D tensor S_v containing vertical spatial structures

$$S_v = \text{Conv}_{1\times1}\left(\text{Cat}\left(\text{AvgPool}_h(F_{\text{in}}), \text{MaxPool}_h(F_{\text{in}})\right)\right). \tag{6}$$

Similarly, a tensor S_h containing horizontal spatial structures can be generated. To model directional spatial dependencies at different scales while maintaining computational efficiency, a 1-D multiscale strip convolution is adopted. Parallel depthwise separable 1-D convolutions with kernel size $k = 7$ and dilation rates $\mathcal{D} = \{1, 2, 3\}$ are applied to S_v. The outputs are fused by elementwise addition and activated by a Sigmoid function to generate the vertical attention weight W_v

$$W_v = \sigma\left(\sum_{d\in\mathcal{D}} \text{DWConv1D}_{k=7,d}(S_v)\right) \tag{7}$$

where $\text{DWConv1D}_{k=7,d}(\cdot)$ denotes a depthwise separable 1-D convolution with kernel size 7 and dilation rate d. Similarly, the horizontal attention weight W_h is obtained by applying strip convolutions along the vertical direction. Since the importance of horizontal and vertical information varies across different scenes, DSFM further introduces a cross-direction gating mechanism to adaptively balance the two directional responses. Channel-level descriptors are obtained through lightweight mapping and global pooling, and directional weights λ_v and λ_h are generated. The direction-enhanced feature is defined as F_{dir}

$$F_{\text{dir}} = \lambda_v\left(S_v \cdot W_v\right) + \lambda_h\left(S_h \cdot W_h\right). \tag{8}$$

To supplement channelwise feature interactions, a lightweight channel attention (CA) module [51] is applied in parallel to produce channel-recalibrated features F_{ca}. The aggregated feature is then obtained through learnable scaling coefficients α and β

$$F_{\text{agg}} = F_{\text{in}} + \alpha F_{\text{dir}} + \beta F_{\text{ca}} \tag{9}$$

where α controls the injection intensity of directional spatial information, and β controls that of channel context. Both are automatically optimized during training, allowing the network to flexibly balance the attention to spatial and channel features according to the needs of different scenes.

Finally, to explicitly focus on change regions, DSFM utilizes the difference information of bitemporal features for gating modulation. Specifically, the absolute difference of bitemporal features is calculated, and a spatial difference gating map D_{diff} is generated through a convolutional layer. The final output feature F_{fuse} of the module is defined as

$$F_{\text{fuse}} = \phi\left(\text{Conv}_{3\times3}\left(F_{\text{agg}} \cdot D_{\text{diff}} + F_{\text{agg}}\right)\right). \tag{10}$$

Through this design, the output of DSFM contains both long-range geometric constraints obtained by DA and change saliency guidance injected by difference gating, thereby providing a more discriminative change feature representation for the subsequent multiscale interaction.

D. Multiscale Interaction Module

In the preceding modules, DWAM and DSFM characterize change cues from different perspectives, yet their operating scales differ. To coordinate frequency and spatial feature responses across multiple spatial scales, this article introduces the MIM to enhance cross-scale semantic consistency. Its structure is shown in Fig. 2.

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MIM adopts a multikernel parallel structure to construct multiscale receptive fields within the same feature level, integrating context information at different spatial scales with low computational cost. Given the input feature F_{fuse} from DSFM, a 1×1 convolution is first applied to project and compress the channel dimension, yielding the intermediate feature representation

$$F_{\text{mid}} = \phi\left(\text{Conv}_{1\times1}(F_{\text{fuse}})\right). \tag{11}$$

Subsequently, to explicitly capture spatial context information at different scales, F_{mid} is fed into a set of parallel depthwise separable convolution branches. Each branch employs a different kernel size $\mathcal{K} = \{3, 5, 7, 9, 11\}$ to model contextual information from local to larger spatial ranges. The outputs of different scale branches are fused via elementwise summation, thereby achieving effective multiscale feature integration while maintaining compact feature dimensions

$$F_{\text{multi}} = \sum_{k\in\mathcal{K}} \text{DWConv}_{k\times k}(F_{\text{mid}}) \tag{12}$$

where $\text{DWConv}_{k\times k}(\cdot)$ denotes the depthwise convolution operation with kernel size k.

Since depthwise convolution performs spatial aggregation independently within each channel, the fused feature F_{multi} is further processed by a pointwise convolution to promote interchannel information interaction and achieve joint modeling of multiscale features. To facilitate effective gradient propagation and preserve original feature information, MIM adopts a residual connection structure. Specifically, the multiscale feature refined by pointwise convolution is added to the original input feature to obtain the final output

$$F_{\text{out}} = F_{\text{fuse}} + \text{Conv}_{1\times1}\left(\phi\left(\text{Conv}_{1\times1}(F_{\text{multi}})\right)\right). \tag{13}$$

By reaggregating directional spatial features across multiple spatial scales, MIM effectively coordinates the scale discrepancies between frequency-domain reliability screening and spatial directional modeling, ensuring that frequency suppression and enhancement effects remain consistent across spatial contexts of different scales. This provides semantically more stable change feature representations for the subsequent decoding stage.

Finally, the multilevel features processed by MIM are fed into a lightweight decoder. The decoder adopts a progressive upsampling strategy with skip connections, fusing deep semantic features and shallow spatial detail features layer by layer, and progressively restoring the spatial resolution of feature maps. The final binary change detection map is generated by a classification head as $P \in \mathbb{R}^{H\times W\times 2}$.

E. Loss Function

For binary change detection, the task can be regarded as a pixelwise binary classification problem. Accordingly, the cross-entropy loss is adopted for model optimization

$$\mathcal{L}_{\text{ce}} = -\frac{1}{H_0 W_0}\sum_{i=1}^{H_0}\sum_{j=1}^{W_0}\left[Y_{i,j}\log P_{i,j} + (1 - Y_{i,j})\log(1 - P_{i,j})\right] \tag{14}$$

where $\mathbf{Y}$ denotes the ground-truth change map, and H_0 and W_0 represent the height and width of the input bitemporal images, respectively.

IV. Experiments

A. Datasets

We evaluate the proposed LDDN on four publicly available RSCD datasets, covering diverse scenes and change types.

1) SYSU-CD [52]: The SYSU-CD dataset consists of 20 000 pairs of very high-resolution satellite images acquired between 2007 and 2014, supporting multiclass change detection scenarios such as building construction, vegetation variation, and marine development. The original images are cropped into patches of size 256×256 pixels and split into training, validation, and test sets with a ratio of 6:2:2, resulting in 12 000, 4000, and 4000 image pairs, respectively. This dataset enables comprehensive evaluation across diverse change patterns.

2) CLCD [53]: The CLCD dataset contains 600 pairs of cultivated land change samples collected by the Chinese Gaofen-2 satellite over Guangdong Province in 2017 and 2019, with spatial resolutions ranging from 0.5 to 2 m. Each sample consists of two 512×512 images and a corresponding binary cropland change label. To facilitate model training, the images are further cropped into patches and randomly divided into training (1440), validation (480), and test (480) sets.

3) WHU-CD [54]: The WHU-CD dataset is a large-scale urban building change detection benchmark composed of co-registered aerial images captured in 2012 and 2016, with an original spatial resolution of 0.075 m per pixel. The original image size is $32\,507 \times 15\,354$ pixels and includes building construction and demolition changes. For efficient training, the images are partitioned into nonoverlapping 256×256 patches, which are randomly assigned to the training (5947), validation (743), and test (744) sets. This preprocessing strategy preserves fine-grained spatial details while improving training efficiency.

4) LEVIR-CD [55]: LEVIR-CD is a benchmark dataset specifically designed for building change detection, containing 637 pairs of images with a spatial resolution of 0.5 m and an image size of 1024×1024 pixels. The dataset covers urban areas in Texas from 2002 to 2018. To adapt to deep learning frameworks and reduce computational overhead, the original images are cropped into nonoverlapping 256×256 patches, resulting in 7120 training pairs, 1024 validation pairs, and 2048 test pairs.

B. Experimental Details

All experiments are implemented using the PyTorch framework. Model training and evaluation are conducted on a workstation equipped with a single NVIDIA A800 GPU. During training, we apply random spatial flipping, where each image pair has a 50% probability of being flipped horizontally and an independent 50% probability of being flipped vertically. In addition, random temporal exchange is adopted by swapping the temporal order of the bitemporal image pairs with a 50% probability. The network parameters

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TABLE I
QUANTITATIVE COMPARISONS ON SYSU-CD AND CLCD. BEST/SECOND-BEST RESULTS ARE MARKED IN RED/BLUE

Methods	Years	SYSU (%)					CLCD (%)				
		F1 ↑	IoU ↑	Pre ↑	Rec ↑	OA ↑	F1 ↑	IoU ↑	Pre ↑	Rec ↑	OA ↑
FC-Diff	2018	67.49±0.31	50.93±0.35	91.66±0.35	53.40±0.28	87.86±0.11	58.68±0.66	41.53±0.66	72.55±2.09	49.30±1.15	94.84±0.12
SNUNet	2021	80.46±0.36	67.31±0.50	81.89±1.09	79.11±1.72	90.94±0.03	69.37±0.81	53.11±0.94	71.47±2.77	67.45±0.99	95.57±0.23
BIT	2021	80.20±0.67	66.95±0.93	83.60±1.00	77.10±1.76	91.03±0.21	73.91±1.41	58.63±1.78	78.48±1.73	69.85±1.38	96.33±0.20
DMINet	2023	82.23±0.52	69.82±0.75	84.96±1.01	79.70±1.82	91.88±0.12	74.39±1.00	59.23±1.27	79.32±1.83	70.06±0.59	96.41±0.17
FIMP	2024	81.37±0.23	68.59±0.32	81.33±2.00	81.48±2.33	91.20±0.19	68.70±1.42	52.34±1.64	67.97±1.60	69.47±1.75	95.29±0.22
WS-Net++	2024	80.14±0.48	66.87±0.67	83.49±0.31	77.06±0.63	91.00±0.20	69.05±1.57	52.75±1.82	72.53±2.04	65.98±3.03	95.60±0.19
S2ENet	2025	83.21±0.67	71.26±0.98	86.12±0.42	80.52±1.61	92.34±0.21	71.28±0.80	55.37±0.97	72.43±0.81	70.20±2.27	95.79±0.04
A2Net	2023	82.55±0.43	70.28±0.63	86.03±1.43	79.36±1.34	92.09±0.21	75.76±0.97	60.99±1.25	77.94±2.68	73.75±0.87	96.49±0.21
RFANet	2024	82.44±0.20	70.12±0.29	84.29±2.27	80.72±1.72	91.89±0.28	74.86±1.23	59.83±1.58	77.18±1.40	72.68±1.30	96.37±0.18
LCDNet	2025	82.66±0.12	70.44±0.18	85.86±0.43	79.69±0.42	92.11±0.06	75.70±0.23	60.90±0.30	79.39±2.24	72.41±2.12	96.54±0.09
Ours	–	83.33±0.28	71.42±0.42	86.20±1.25	80.66±1.04	92.39±0.17	76.59±0.39	62.06±0.51	79.64±1.01	73.77±0.28	96.64±0.08

TABLE II
QUANTITATIVE COMPARISONS ON WHU-CD AND LEVIR-CD. BEST/SECOND-BEST RESULTS ARE MARKED IN RED/BLUE

Methods	Years	WHU (%)					LEVIR (%)				
		F1 ↑	IoU ↑	Pre ↑	Rec ↑	OA ↑	F1 ↑	IoU ↑	Pre ↑	Rec ↑	OA ↑
FC-Diff	2018	82.75±0.13	70.57±0.19	89.57±0.48	76.90±0.57	98.73±0.01	87.45±0.57	77.70±0.90	90.04±0.92	85.02±1.12	98.76±0.05
SNUNet	2021	92.40±0.14	85.88±0.25	93.82±0.40	91.03±0.44	99.41±0.01	90.65±0.17	82.89±0.28	92.51±0.17	88.86±0.27	99.07±0.02
BIT	2021	93.04±0.22	86.99±0.38	94.55±0.30	91.58±0.26	99.46±0.02	90.49±0.09	82.63±0.16	91.50±0.75	89.51±0.84	99.04±0.01
DMINet	2023	93.58±0.18	87.94±0.31	94.21±0.27	92.97±0.14	99.49±0.01	90.85±0.23	83.24±0.39	92.51±0.09	89.26±0.53	99.08±0.02
FIMP	2024	90.83±1.17	83.21±1.97	92.84±1.12	88.95±2.69	99.29±0.08	90.90±0.41	83.32±0.69	92.36±0.27	89.49±0.71	99.09±0.04
WS-Net++	2024	91.25±0.24	83.90±0.40	93.01±0.56	89.56±0.90	99.32±0.01	89.88±0.15	81.61±0.25	91.44±0.13	88.37±0.41	98.99±0.01
S2ENet	2025	93.45±0.15	87.70±0.26	94.45±0.67	92.47±0.41	99.49±0.01	90.97±0.20	83.44±0.34	91.76±0.43	90.20±0.65	99.09±0.02
A2Net	2023	94.25±0.46	89.13±0.83	96.12±0.34	92.46±0.58	99.55±0.04	91.35±0.14	84.08±0.24	92.67±0.28	90.07±0.51	99.13±0.01
RFANet	2024	94.32±0.29	89.24±0.53	95.35±0.78	93.31±0.62	99.55±0.02	91.48±0.04	84.29±0.06	92.50±0.25	90.47±0.31	99.14±0.01
LCDNet	2025	93.76±0.41	88.25±0.73	95.06±0.55	92.50±0.29	99.51±0.03	91.04±0.04	83.56±0.06	92.61±0.46	89.53±0.36	99.10±0.01
Ours	–	95.11±0.22	90.67±0.39	96.28±0.60	93.96±0.15	99.62±0.02	91.65±0.08	84.33±0.23	92.92±0.12	90.40±0.04	99.16±0.01

are optimized using the AdamW optimizer with a weight decay of 0.01. The initial learning rate is set to 0.001 and scheduled using a cosine annealing strategy, with the minimum learning rate set to 0.0001. The model is trained for 150 epochs with a batch size of 32. All experiments were repeated over three independent runs. For the main quantitative comparison and module ablation experiments, results are reported as the mean ± standard deviation, while the backbone-selection experiment reports the mean values for compact presentation. For quantitative evaluation, five commonly used metrics in change detection are adopted, including overall accuracy (OA), precision (Pre), recall (Rec), $F1$-score ($F1$), and intersection over union (IoU) [56].

C. Comparison With Sota Methods

To comprehensively evaluate the effectiveness and superiority of the proposed LDDN, we compare it with ten representative state-of-the-art change detection methods: FC-Diff [15], SNUNet [37], BIT [20], DMINet [36], FIMP [35], WS-Net++ [34], S2ENet [33], A2Net [27], RFANet [29], and LCDNet [28]. These methods cover classical deep learning baselines, frequency-domain-aware networks, and lightweight architectures. To ensure fair comparison, all competing methods are retrained and evaluated under the same experimental settings using the official implementations provided by the authors.

Among them, FC-Diff represents an early FCN for change detection, employing a Siamese encoder to extract features and multilevel difference maps for prediction. SNUNet introduces a densely connected Siamese architecture to alleviate information loss in deep features and enhance localization accuracy. BIT is a pioneering work that introduces Transformer architectures into change detection, leveraging token-based self-attention to model long-range spatiotemporal dependencies. FIMP proposes a Fourier feature interaction module to mitigate imaging style discrepancies by adaptively filtering bitemporal features in the frequency domain. WS-Net++ is a wavelet-based Siamese network that enhances change boundary completeness through spatial–frequency interaction and further incorporates semi-supervised domain adaptation to reduce appearance shifts. S2ENet is a structure-sensitive network that analyzes high-frequency subbands in shallow features to adaptively enhance edge structures while suppressing high-frequency pseudo-changes via denoising mechanisms. A2Net targets the limited representational capacity of mobile networks by introducing neighborhood aggregation and progressive change recognition modules, achieving effective feature aggregation with extremely low parameter counts. RFANet proposes a robust feature aggregation network that promotes dense interlevel interaction through feature enhancement modules and captures global semantics via a semantic segmentation–aggregation strategy. LCDNet adopts MobileNetV2 as the backbone and combines temporal interaction with gating mechanisms to dynamically adjust channel weights, aiming to balance computational cost and detection accuracy in resource-constrained environments.

1) Quantitative Evaluation: Comprehensive quantitative comparisons on the SYSU-CD, CLCD, WHU-CD, and LEVIR-CD datasets are reported in Tables I and II. Overall, LDDN achieves the best $F1$-score and IoU on all

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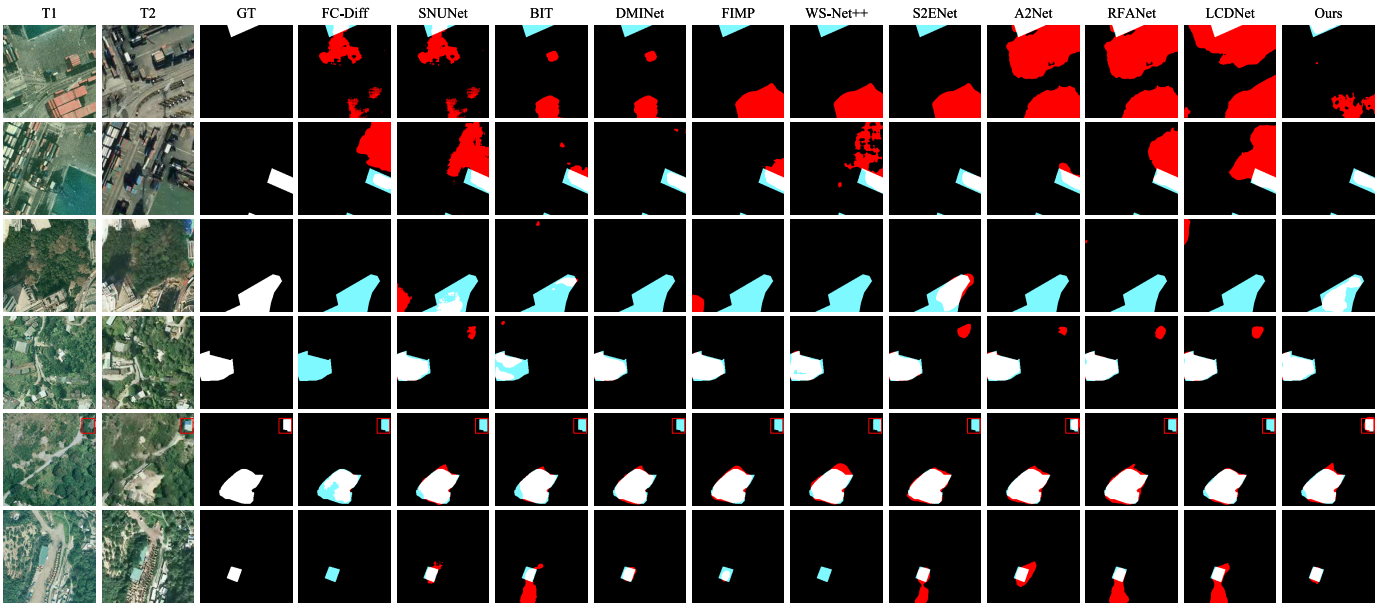


Fig. 4. Visualization results on the SYSU-CD dataset. T1 and T2 are bitemporal images, and GT is the ground truth. White, black, red, and blue are used for TP, TN, FP, and FN, respectively.

TABLE III
COMPARISON OF FLOPS AND PARAMETERS BETWEEN THE PROPOSED METHOD AND TEN COMPARISON METHODS

Methods	Backbone	Params (M)↓	FLOPs (G)↓
FC-Diff	w/o	1.35	4.73
SNUNet	w/o	12.03	54.83
BIT	ResNet18	3.01	8.5
DMINet	ResNet18	6.24	14.55
FIMP	EfficientNet-B4	2.45	2.85
WS-Net++	ResNet18	237.54	33.82
S2ENet	ResNet18	17.59	24.92
A2Net	MobileNetV2	3.78	3.05
RFANet	MobileNetV2	2.86	3.16
LCDNet	MobileNetV2	2.56	4.45
Ours	EfficientNet-B4	1.98	2.58

Note: The abbreviation "w/o" indicates that a pre-trained backbone was not used and was instead replaced by the modules designed in the respective papers.

four datasets, demonstrating its stable and competitive performance.

On SYSU-CD, LDDN achieves the highest $F1$-score of 83.33% and IoU of 71.42%. Although FIMP and RFANet obtain higher recall scores, their precision scores are lower than those of LDDN. This may be because FIMP's Fourier-domain interaction and RFANet's feature aggregation tend to preserve more potential change candidates, but may also introduce more false positives in complex scenes. In contrast, LDDN achieves a better precision–recall balance through frequency-domain reliability screening and spatial directional modeling. S2ENet also performs competitively, likely benefiting from its frequency–spatial collaborative strategy, but it requires higher computational complexity, as shown in Table III. On the challenging CLCD dataset, LDDN achieves the best results across all evaluation metrics, outperforming A2Net by 0.83% and 1.07% in $F1$-score and IoU, respectively. Since cropland changes often involve elongated field boundaries, ridges, and irregular structural variations, the directional dependency modeling in DSFM is well-suited to capturing such anisotropic change patterns, while DWAM helps suppress unreliable responses caused by local texture interference. This explains the stable improvement of LDDN in complex cropland scenes. Similarly, on WHU-CD, the proposed method achieves the best results across all metrics. Notably, it reaches an $F1$-score of 95.11% and an IoU of 90.67%, representing improvements of 0.79% and 1.43% over the second-best method, RFANet. These results suggest that the joint modeling of response reliability and structural continuity is beneficial for accurately detecting building changes with clear geometric boundaries. For the LEVIR-CD dataset, LDDN maintains its lead in comprehensive accuracy, yielding the highest $F1$-score and precision. Although RFANet obtains a slightly higher recall of 90.47% compared with 90.40% for LDDN, its lower precision indicates a higher tendency toward false alarms. The higher precision of LDDN can be attributed to DWAM, which adaptively screens unreliable directional high-frequency responses and reduces pseudo-change interference. Overall, these results show that the proposed dual-domain design improves change detection not by simply increasing change responses, but by achieving a more balanced tradeoff between false-alarm suppression and true-change preservation.

To further evaluate model size and computational complexity, Table III presents the parameter counts and FLOPs of LDDN and ten competing methods. Since different methods employ different backbones or customized encoders, these complexity statistics should be interpreted as an end-to-end comparison of complete change detection systems from a deployment perspective, rather than as a strictly backbone-controlled fairness comparison. Under this practical setting, LDDN maintains low computational cost

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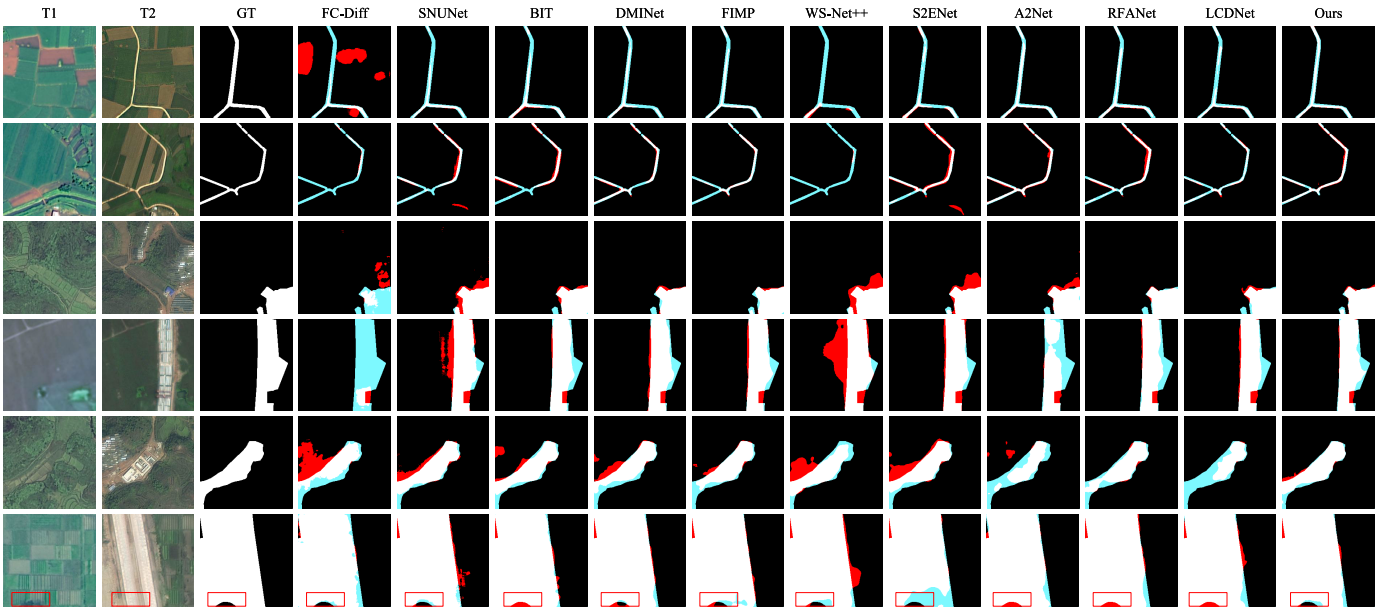


Fig. 5. Visualization results on the CLCD dataset. T1 and T2 are bitemporal images, and GT is the ground truth. White, black, red, and blue are used for TP, TN, FP, and FN, respectively.

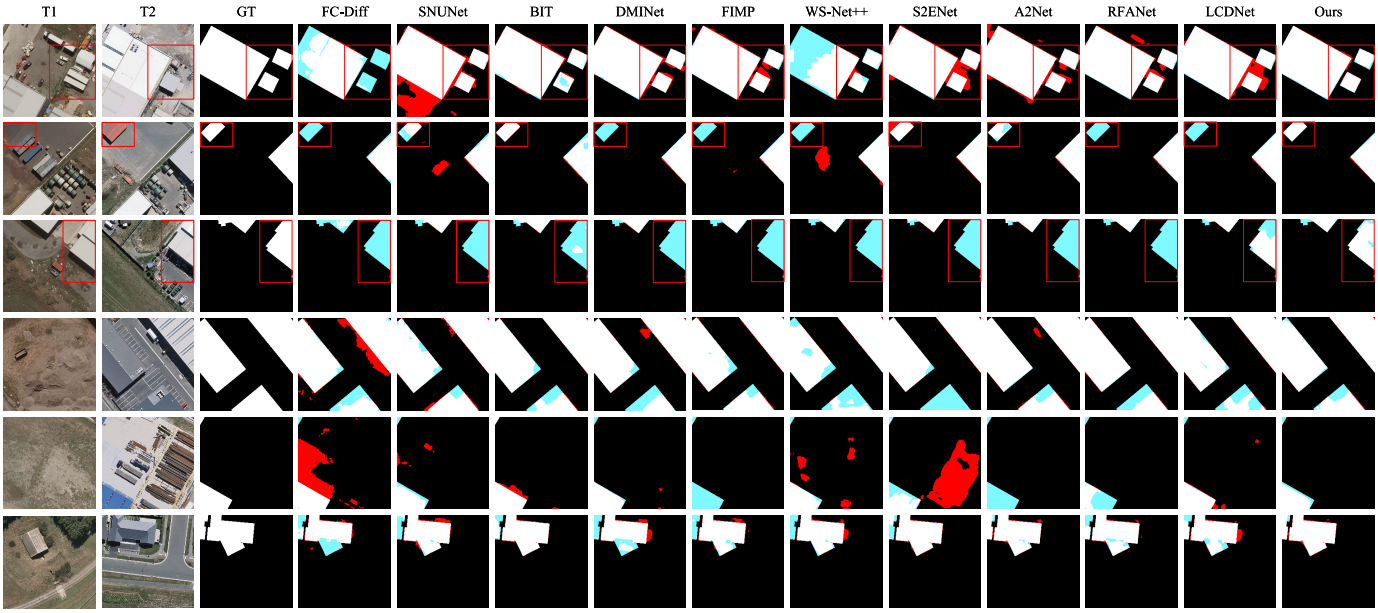


Fig. 6. Visualization results on the WHU-CD dataset. T1 and T2 are bitemporal images, and GT is the ground truth. White, black, red, and blue are used for TP, TN, FP, and FN, respectively.

while achieving competitive or superior overall detection performance.

2) Qualitative Evaluation: Figs. 4–7 present the qualitative comparison results on the SYSU-CD, CLCD, WHU-CD, and LEVIR-CD datasets, with particular emphasis on challenging scenarios involving pseudo-changes, weak appearance contrast, and ambiguous boundaries. These examples illustrate the differences between LDDN and existing methods.

a) Robustness to pseudo-changes: Local textural noise and temporary object interference are the primary causes of false alarms. As shown in the 1st and 2nd rows of Fig. 4, the 3rd and 4th rows of Fig. 5, and the 4th and 5th rows of Fig. 7, several comparison methods produce obvious false alarms in unchanged regions. In particular, some methods erroneously label vehicles, containers, or nonbuilding areas as changes. This may be because methods mainly relying on spatial-domain appearance modeling tend to be sensitive to local color, texture, or illumination differences while lacking explicit reliability screening for high-frequency pseudo-change responses. Although frequency-aware methods can alleviate this problem to some extent, global or insufficiently directional frequency filtering may still retain unreliable local responses. In contrast, LDDN produces cleaner change maps with fewer false positives. This advantage mainly comes from DWAM,

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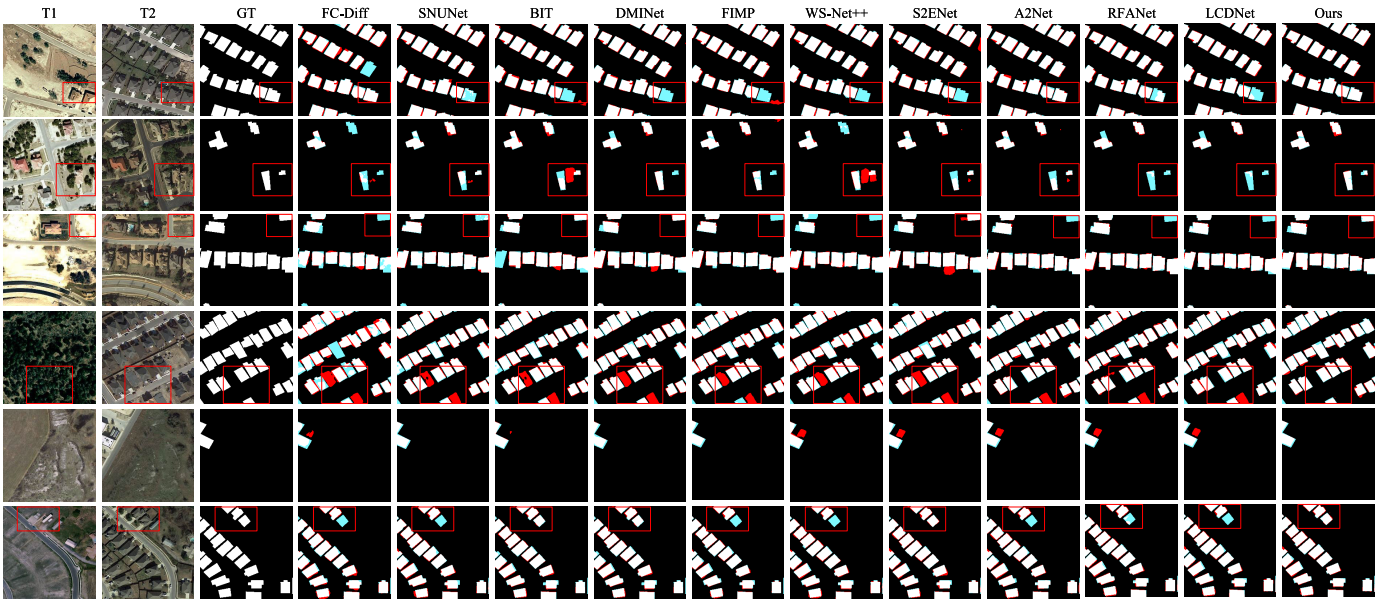


Fig. 7. Visualization results on the LEVIR-CD dataset. T1 and T2 are bitemporal images, and GT is the ground truth. White, black, red, and blue are used for TP, TN, FP, and FN, respectively.

TABLE IV
ABLATION STUDY RESULTS OF DWAM, DSFM, AND MIM IN LDDN ON SYSU-CD AND CLCD

DWAM	DSFM	MIM	Params (M)↓	FLOPs (G)↓	SYSU (%) F1↑	SYSU (%) IoU↑	SYSU (%) Pre↑	SYSU (%) Rec↑	SYSU (%) OA↑	CLCD (%) F1↑	CLCD (%) IoU↑	CLCD (%) Pre↑	CLCD (%) Rec↑	CLCD (%) OA↑
			1.46	1.99	81.06±0.33	68.15±0.47	83.51±2.04	78.82±2.47	91.32±0.08	74.04±0.46	58.78±0.59	76.33±2.38	72.00±2.79	96.24±0.08
✓			1.50	2.02	81.42±0.36	68.67±0.51	85.64±0.44	77.61±0.96	91.65±0.10	74.49±0.16	59.35±0.20	76.83±0.72	72.30±0.94	96.32±0.02
	✓		1.83	2.38	82.82±0.06	70.68±0.08	83.90±0.57	**81.77±0.63**	92.00±0.04	75.51±0.32	60.66±0.41	76.96±0.19	**74.12±0.79**	96.42±0.02
		✓	1.57	2.68	82.07±0.52	69.60±0.75	84.48±1.43	79.83±1.29	91.78±0.27	73.39±0.37	57.97±0.46	75.84±3.13	71.24±2.73	96.16±0.17
✓	✓		1.87	2.39	82.48±0.11	70.19±0.16	85.62±1.02	79.58±0.93	92.03±0.09	75.67±0.98	60.87±1.26	78.50±2.15	73.06±0.45	96.50±0.19
✓		✓	1.61	2.20	81.97±0.32	69.44±0.46	84.63±0.88	79.48±1.26	91.75±0.08	75.77±0.60	61.00±0.79	78.42±1.18	73.32±1.53	96.51±0.08
	✓	✓	1.94	2.56	82.76±0.33	70.59±0.47	85.07±1.02	80.58±1.12	92.08±0.15	75.16±1.46	60.22±1.89	78.19±2.06	72.38±1.78	96.44±0.22
✓	✓	✓	1.98	2.58	**83.33±0.28**	**71.42±0.42**	**86.20±1.25**	80.66±1.04	**92.39±0.17**	**76.59±0.39**	**62.06±0.51**	**79.64±1.01**	73.77±0.28	**96.64±0.08**

which adaptively screens directional high-frequency subbands and suppresses unreliable responses while preserving structural cues associated with real changes.

b) Sensitivity to weak changes: Missed detections easily occur when the spectral characteristics of change targets are extremely similar to the background. As observed in the complex urban scenes in the 3rd row of Fig. 6 and the 1st and 3rd rows of Fig. 7, several comparison methods exhibit extensive missed detections due to the similarity between building shadows or roof materials and the background. In these cases, the change regions are not accompanied by strong appearance contrast, and their responses may be weakened during feature extraction and fusion. This makes it difficult for comparison methods to distinguish weak but semantically meaningful changes from background regions. By contrast, LDDN achieves more complete detection of low-contrast change targets. This improvement can be attributed to the long-range directional constraints introduced by DSFM, which help aggregate contextual evidence along horizontal and vertical directions, and the multiscale perception of MIM, which further complements weak responses across different receptive fields.

c) Preservation of structural continuity: For remote sensing objects with clear directional or elongated structures, such as farmland boundaries, roads, and building contours, maintaining boundary continuity is essential. As shown in the first two rows of Fig. 5 and the representative building cases in Fig. 6, several comparison methods produce fragmented or blurred predictions along object boundaries. Different from weak-change cases, where the main difficulty lies in low appearance contrast, these examples mainly reflect the challenge of preserving continuous geometric structures. Methods dominated by local or isotropic spatial aggregation may capture partial boundary responses but struggle to maintain long-range directional consistency, especially for elongated objects with large aspect ratios. By contrast, the proposed DSFM explicitly models horizontal and vertical dependencies through directional strip operations, while DWAM suppresses unreliable high-frequency interference before spatial fusion. Therefore, LDDN better preserves the structural continuity of elongated change regions and produces more complete boundaries.

D. Ablation Study

To verify the effectiveness of each core component in LDDN, we conduct extensive ablation experiments on the SYSU-CD, CLCD, WHU-CD, and LEVIR-CD datasets. The experimental results are summarized in Tables IV and V.

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TABLE V
ABLATION STUDY RESULTS OF DWAM, DSFM, AND MIM IN LDDN ON WHU-CD AND LEVIR-CD

DWAM	DSFM	MIM	Params (M)↓	FLOPs (G)↓	WHU (%) F1↑	WHU (%) IoU↑	WHU (%) Pre↑	WHU (%) Rec↑	WHU (%) OA↑	LEVIR (%) F1↑	LEVIR (%) IoU↑	LEVIR (%) Pre↑	LEVIR (%) Rec↑	LEVIR (%) OA↑
			1.46	1.99	94.40±0.20	89.40±0.37	95.20±0.44	93.62±0.04	99.56±0.02	91.14±0.15	83.72±0.25	92.22±0.14	89.95±0.54	99.11±0.01
✓			1.50	2.02	94.77±0.02	90.06±0.04	95.86±0.06	93.71±0.09	99.59±0.01	91.19±0.02	83.80±0.04	92.36±0.37	90.18±0.15	99.11±0.01
	✓		1.83	2.38	94.97±0.02	90.43±0.03	96.22±0.35	93.76±0.35	99.61±0.01	91.46±0.10	84.27±0.16	92.31±0.33	**90.63±0.45**	99.14±0.01
		✓	1.57	2.68	94.91±0.16	90.31±0.29	96.23±0.27	93.62±0.06	99.60±0.01	91.42±0.07	84.19±0.11	92.43±0.23	90.42±0.34	99.13±0.01
✓	✓		1.87	2.39	94.63±0.27	89.81±0.48	95.47±0.17	93.81±0.55	99.58±0.02	91.39±0.07	84.14±0.13	92.81±0.11	90.01±0.14	99.14±0.01
✓		✓	1.61	2.20	94.83±0.29	90.16±0.53	96.26±0.31	93.37±0.28	99.60±0.02	91.35±0.07	84.08±0.12	92.76±0.23	89.99±0.21	99.13±0.01
	✓	✓	1.94	2.56	95.01±0.19	90.49±0.34	95.92±0.29	**94.11±0.53**	99.61±0.01	91.42±0.01	84.20±0.02	92.76±0.25	90.12±0.21	99.14±0.01
✓	✓	✓	1.98	2.58	**95.11±0.22**	**90.67±0.39**	**96.28±0.60**	93.96±0.15	**99.62±0.02**	**91.65±0.08**	**84.33±0.23**	**92.92±0.12**	90.40±0.04	**99.16±0.01**

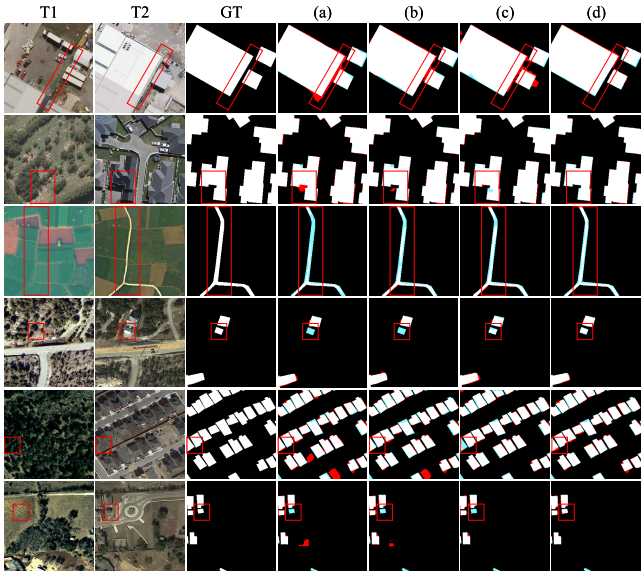


Fig. 8. Ablation visualization of different module combinations. (a)–(d) Columns denote the baseline, baseline + DWAM, baseline + DWAM + DSFM, and full LDDN, respectively. White, black, red, and blue indicate TP, TN, FP, and FN, respectively.

1) Effectiveness of DWAM: DWAM aims to enhance the reliability of bitemporal features by adaptively screening directional high-frequency responses in the wavelet domain. As shown in the first and second rows of Tables IV and V, introducing DWAM into the baseline brings consistent but moderate improvements across the four datasets. Specifically, the $F1$-scores on SYSU-CD, CLCD, WHU-CD, and LEVIR-CD increase by 0.36%, 0.45%, 0.37%, and 0.05%, respectively, while the IoU scores improve by 0.52%, 0.57%, 0.66%, and 0.08%, respectively. More importantly, DWAM consistently improves precision, with gains of 2.13%, 0.50%, 0.66%, and 0.14% on the four datasets. These results indicate that DWAM mainly contributes to reducing false alarms by enhancing the reliability of change responses.

To provide a more intuitive understanding of the contribution of each module, we select representative samples covering typical challenging cases, including pseudo-changes, weak responses, and fragmented structures. As shown in Fig. 8, the prediction results of different module combinations are visualized to illustrate how the proposed modules progressively reduce false positives and false negatives. Compared with the baseline in column Fig. 8(a), the predictions after introducing DWAM in column Fig. 8(b) contain fewer red false-positive regions, especially in the first and second rows,

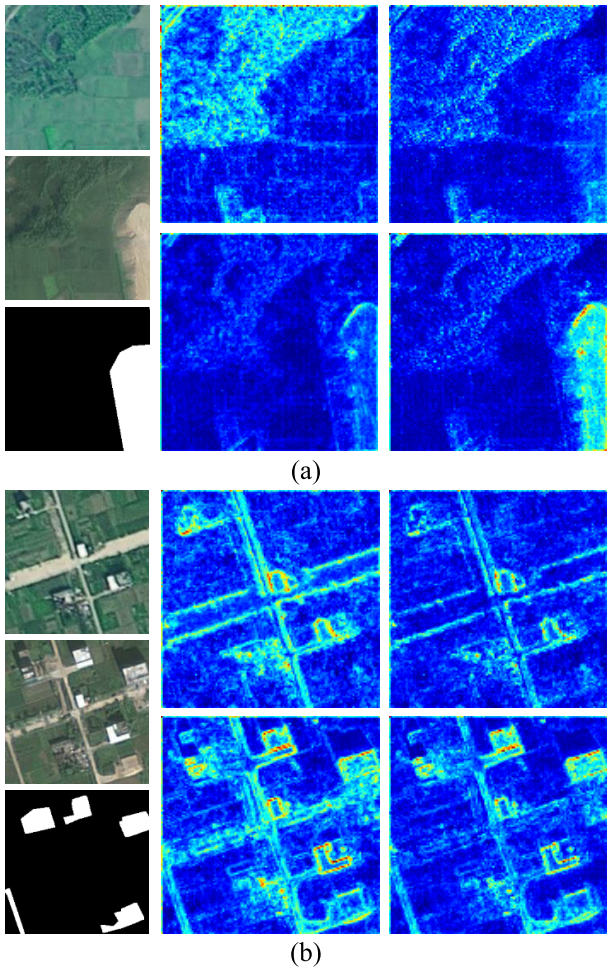


Fig. 9. Visualization of feature maps without and with applying DWAM. (Top to bottom) (a) and (b) First column shows the T1 image, the T2 image, and the ground truth, respectively; the second column presents the bitemporal feature maps without DWAM; and the third column presents the feature maps with DWAM.

where local texture noise and illumination variations cause pseudo-change interference. In addition, the feature visualization in Fig. 9 shows that DWAM suppresses chaotic activations in nonchanging background regions and produces cleaner feature responses. These observations are consistent with the quantitative precision improvements and suggest that DWAM effectively attenuates unreliable high-frequency responses, providing a more stable feature foundation for subsequent spatial modeling.

2) Effectiveness of DSFM: DSFM is designed to enhance spatial discriminability by decoupling 2-D spatial context

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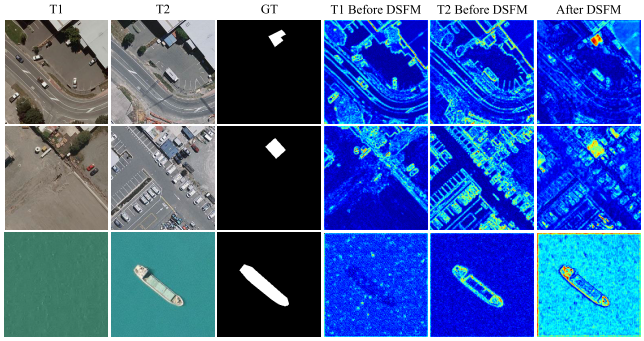


Fig. 10. Visualization of feature maps before and after applying DSFM. T1 and T2 denote the bitemporal images, and GT denotes the ground truth. "T1 before DSFM" and "T2 before DSFM" indicate the feature maps of T1 and T2 before being input into DSFM, while "after DSFM" shows the feature maps after DSFM processing.

into horizontal and vertical directional dependencies. As shown in the first and third rows of Tables IV and V, introducing DSFM alone consistently improves the baseline performance on all four datasets. Specifically, the $F1$-scores on SYSU-CD, CLCD, WHU-CD, and LEVIR-CD increase by 1.76%, 1.47%, 0.57%, and 0.32%, respectively, while the IoU scores improve by 2.53%, 1.88%, 1.03%, and 0.55%, respectively. More importantly, DSFM brings clear recall gains of 2.95%, 2.12%, 0.14%, and 0.68% on the four datasets, indicating that directional spatial modeling mainly contributes to reducing missed detections, especially on challenging datasets with complex scenes and weak change responses.

The visual results in Fig. 8 further support this observation. In particular, the third and fourth rows are selected to highlight weak-change regions and thin elongated structures, such as farmland roads and low-contrast buildings. Compared with the results before applying DSFM in column Fig. 8(b), the predictions after introducing DSFM in column Fig. 8(c) contain fewer blue false-negative regions, showing that DSFM can recover weak or fragmented change targets more effectively. In addition, the feature visualization in Fig. 10 shows that DSFM enhances the structural saliency of true change regions, which further verifies its role in improving spatial discrimination.

3) Effectiveness of MIM: MIM is designed to coordinate the frequency-domain reliability screening of DWAM and the spatial directional modeling of DSFM across multiple scales. Instead of introducing a separate change cue, MIM further refines the dual-domain features by aggregating contextual information with different receptive fields, thereby improving semantic consistency and structural completeness. As shown in the 5th and 8th rows of Tables IV and V, adding MIM to the dual-domain baseline composed of DWAM and DSFM consistently improves the performance on all four datasets. Specifically, the $F1$-scores on SYSU-CD, CLCD, WHU-CD, and LEVIR-CD increase by 0.85%, 0.92%, 0.48%, and 0.26%, respectively, while the IoU scores improve by 1.23%, 1.19%, 0.86%, and 0.19%, respectively. These improvements indicate that MIM effectively enhances the overall robustness of LDDN by compensating for residual semantic inconsistency after frequency-domain filtering and spatial directional fusion.

The visual results in Fig. 8 further demonstrate the refinement effect of MIM. The 5th and 6th rows are selected to show challenging cases with residual missed detections and multiscale change regions. Although the combination of DWAM and DSFM in column Fig. 8(c) can capture the main change regions, some small or fragmented structures remain incomplete. After introducing MIM, the full model in column Fig. 8(d) produces more complete change maps with fewer blue false-negative regions and fewer red false-positive regions. This suggests that MIM helps recover missing details while preserving the noise suppression effect of DWAM, leading to more coherent and reliable predictions.

4) Synergistic Effect and Discussion: The ablation results indicate that the proposed modules in LDDN play complementary roles rather than contributing in a simple additive manner. DWAM and DSFM exhibit different optimization tendencies. Specifically, DWAM mainly improves prediction reliability by suppressing unreliable high-frequency responses, which generally benefits precision, whereas DSFM enhances spatial discriminability by modeling directional context and recovering weak or elongated change regions, which is more closely related to recall improvement. However, directly combining these two modules may introduce a potential imbalance between frequency-domain filtering and spatial-domain enhancement. For example, on the SYSU-CD dataset, adding DWAM to DSFM increases precision from 83.90% to 85.62%, but decreases recall from 81.77% to 79.58%. This phenomenon suggests that although DWAM effectively suppresses false alarms, overly strong high-frequency filtering may also weaken certain fine-grained or low-contrast change responses that DSFM aims to preserve.

The introduction of MIM helps mitigate this imbalance by coordinating dual-domain features under multiple receptive fields. Instead of introducing an additional type of change cue, MIM reaggregates the features enhanced by DWAM and DSFM at different spatial scales, thereby improving the consistency between false-alarm-suppressing frequency filtering and weak-change-preserving directional spatial modeling. In this way, MIM can alleviate the attenuation of weak structural responses while maintaining the false-alarm suppression effect introduced by DWAM. As a result, the complete LDDN achieves a more favorable balance between precision and recall under a lightweight computational budget. These observations further support the design rationale of combining frequency-domain reliability screening, spatial-domain directional modeling, and multiscale feature interaction within a unified framework.

5) Effectiveness of Backbone Selection: To evaluate the influence of backbone choice within LDDN, we compare three representative lightweight backbones, including ShuffleNet [57], MobileNetV2 [31], and EfficientNet-B4 [49], as shown in Table VI. This experiment aims to analyze the backbone sensitivity of the proposed framework rather than provide a direct fairness comparison with other methods. The results show that ShuffleNet and MobileNetV2 reduce computational cost but suffer from clear performance degradation due to limited feature representation capacity. In contrast, EfficientNet-B4 achieves the best performance, with an $F1$-score of 95.11% and an IoU of 90.67% while maintaining a compact model size of 1.98M parameters and 2.58G FLOPs.

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TABLE VI
QUANTITATIVE COMPARISON OF THE PROPOSED METHOD WITH DIFFERENT BACKBONES ON THE WHU-CD DATASET

Backbone	Params (M)↓	FLOPs (G)↓	WHU (%) F1↑	IoU↑	Pre↑	Rec↑	OA↑
ShuffleNet	1.84	0.47	92.79	86.55	94.05	91.57	99.44
MobileNetV2	1.81	1.05	94.02	88.71	94.86	93.19	99.53
EfficientNet-B4	1.98	2.58	**95.11**	**90.67**	**96.28**	**93.96**	**99.62**

TABLE VII
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE WHU-CD DATASET FOR REGISTRATION ERROR EXPERIMENTS

Method	WHU (%) F1↑	IoU↑	Pre↑	Rec↑	OA↑
FIMP	89.38±0.18	80.81±0.30	88.50±0.37	90.29±0.07	99.15±0.02
WS-Net++	89.23±0.50	80.56±0.82	89.44±0.34	89.02±0.81	99.15±0.04
S2ENet	93.11±0.10	87.11±0.17	93.44±0.10	92.78±0.10	99.46±0.01
A2Net	94.01±0.09	88.70±0.16	95.17±0.08	92.89±0.10	99.53±0.01
RFANet	93.60±0.29	87.96±0.51	93.73±0.60	93.47±0.05	99.49±0.02
LCDNet	92.58±0.34	86.19±0.58	92.83±0.61	92.34±0.12	99.41±0.03
Ours*	93.91±0.30	88.52±0.53	94.01±0.63	**93.82±0.12**	99.52±0.03
Ours	**94.48±0.15**	**89.55±0.28**	**95.89±0.31**	93.12±0.01	**99.57±0.01**

Note: Ours* denotes LDDN without the DWAM module.

Therefore, EfficientNet is adopted as the final backbone, as it provides a better tradeoff between detection accuracy and computational efficiency.

E. Discussion on Misregistration

To further examine the stability of different methods under slight spatial inconsistency, an additional registration-error simulation experiment was conducted on the WHU-CD dataset during the testing stage [39]. Specifically, an unregistered state was simulated by randomly perturbing the four vertices of the aligned T2 image and warping the whole image with the corresponding perspective transformation while keeping T1 unchanged. To focus on slight registration errors, the perturbation radius was set to 5 pixels.

Table VII presents the quantitative evaluation of the compared methods under simulated registration errors. As expected, the introduction of slight spatial inconsistencies degrades the detection accuracy across all evaluated models, reflecting the inherent sensitivity of lightweight architectures to geometric misalignment. Nevertheless, the proposed LDDN exhibits a resilient performance profile, maintaining an average $F1$-score of 94.48% and an IoU of 89.55%. Compared with existing lightweight architectures such as A2Net and RFANet, LDDN maintains advantages in $F1$-score, IoU, precision, and OA under misaligned conditions, although its recall is slightly lower than that of RFANet. Furthermore, an ablation comparison between the complete LDDN and its baseline variant without the DWAM validates the effectiveness of DWAM in mitigating misalignment-induced interferences. Specifically, the integration of DWAM yields a notable improvement in precision from 94.01% to 95.89% and reduces the performance variance across multiple independent runs, as evidenced by the standard deviation of the $F1$-score decreasing from 0.30 to 0.15. Since DWAM is primarily designed to suppress high-frequency noise rather than perform explicit spatial alignment, it operates by adaptively filtering out the spurious signals triggered by slight pixel offsets. This targeted suppression mechanism effectively restrains false alarms, thereby providing more reliable change discrimination in scenes with minor registration errors.

V. CONCLUSION

To address the issue of insufficient reliability of lightweight RSCD models in complex high-resolution images under resource-constrained scenarios, this article proposes an LDDN. Taking directional perception as a unified modeling cue, this method effectively improves the stability of change discrimination by uniting frequency-domain denoising and spatial-domain directional modeling under strict computational constraints. Specifically, the DWAM adaptively screens directional high-frequency subbands in the frequency domain, thereby suppressing unreliable high-frequency responses and mining structural cues of real changes. The DSFM enhances the discriminative ability of the lightweight model for weak change regions at a low computational cost. Furthermore, the MIM coordinates dual-domain feature representations across different spatial scales. Extensive experiments conducted on four public benchmarks demonstrate that LDDN achieves a good balance between accuracy and efficiency. With only 1.98M parameters and 2.58G FLOPs, the proposed method achieves competitive or superior performance in comparison with existing lightweight methods and multiple high-performance models. Overall, this study highlights the effectiveness of reliability-aware dual-domain modeling in efficient and robust RSCD. Nevertheless, since LDDN does not explicitly perform geometric alignment, it may still be affected by severe misregistration. Future work will explore registration-aware and label-efficient learning to further improve robustness and reduce annotation dependence.

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